Show that the dot product and cross product is distributive

Solution:

Let the three vectors be

$$\boldsymbol{v}_1 = [a, b, c]$$
$$\boldsymbol{v}_2 = [d, e, f]$$
$$\boldsymbol{v}_3 = [g, h, i]$$

dot product:

Show

$$\boldsymbol{v}_1 \cdot (\boldsymbol{v}_2 + \boldsymbol{v}_3) = \boldsymbol{v_1} \cdot \boldsymbol{v}_2 + \boldsymbol{v_1} \cdot \boldsymbol{v}_3$$

Using

$$\boldsymbol{v}_2 + \boldsymbol{v}_3 = [d + g, e + h, f + i]$$
$$\boldsymbol{v}_1 \cdot \boldsymbol{v}_2 = ad + be + cf$$
$$\boldsymbol{v}_1 \cdot \boldsymbol{v}_3 = ag + bh + ci$$

LHS:

$$\boldsymbol{v}_1 \cdot (\boldsymbol{v}_2 + \boldsymbol{v}_3) = [a, b, c] \cdot [d + g, e + h, f + i] =$$
$$= a(d + g) + b(e + h) + c(f + i)$$

RHS:

$$\boldsymbol{v}_1 \cdot \boldsymbol{v}_2 + \boldsymbol{v}_1 + \boldsymbol{v}_3 = ad + be + cf + ag + bh + ci =$$
$$= a(d + g) + b(e + h) + c(f + i)$$

cross product: Show

$$\boldsymbol{v}_1 \times (\boldsymbol{v}_2 + \boldsymbol{v}_3) = (\boldsymbol{v}_1 \times \boldsymbol{v}_2) + (\boldsymbol{v}_1 \times \boldsymbol{v}_3)$$

Using

$$\boldsymbol{v}_2 + \boldsymbol{v}_3 = [d + g, e + h, f + i]$$
$$\boldsymbol{v}_1 \times \boldsymbol{v}_2 = bf - ce + cd - af + ae - bd$$
$$\boldsymbol{v}_1 \times \boldsymbol{v}_3 = bi - ch + cg - ai + ah - bg$$

which gives

$$(\boldsymbol{v}_1 \times \boldsymbol{v}_2) + (\boldsymbol{v}_1 \times \boldsymbol{v}_3) = a(e + h - f - i) + b(f + i - d - g) + c(d + g - e - h)$$

and LHS:

$$\boldsymbol{v}_1 \times (\boldsymbol{v}_2 + \boldsymbol{v}_3) = b(f + i) - c(e + h) + c(d + g) - a(f + i) + a(e + h) - b(d + g) =$$
$$= bf + bi - ce - ch + cd + cg - af - ai + ae + ah - bd - bg =$$
$$= a(e + h - f - i) + b(f + i - d - g) + c(d + g - e - h)$$

$\square$